

Improving Forest Carbon Models under Climate Change Using Drones and Experiments

Context: There is increasing evidence that anthropogenic climate change, and particularly increased temperature, affects many natural systems at the global scale^{1,2}. The most frequently observed biological impact of climate change over the past decades are major changes on spring and autumn phenology — the timing of recurring life history events^{3,4,5,6,7}. Understanding the consequences of these shifts on ecosystems requires understanding how much the growing season has changed⁸. Spring phenological events (e.g. budburst and leafout) have been advancing from 0.5⁹ to 4.2 days/decade^{10,11} and are mainly driven by temperature^{12,4,13}. In contrast, autumn phenology (e.g. budset and leaf colouring) is delayed, though to a much lesser extent than spring^{14,15}, and is driven by shortening photoperiod^{16,17,18} and colder temperatures^{16,19}. These shifts support a long-lasting and intuitive assumption that earlier spring and delayed autumn events lead to longer seasons—and thus increased growth²⁰. However, research from the past three years has cast doubt on this hypothesis^{21,22,23}. Recently, Dow *et al.* (2022) showed that despite an earlier growth onset, neither growth rate nor overall annual increment was increased by longer seasons. This could substantially affect carbon-cycle model projections and thus feedbacks to future climate^{24,25}. Understanding these findings requires answering why trees do not grow more despite longer growing seasons. I hypothesize two possible drivers explaining this phenomenon: external (environmental)²⁶ or internal (via physiological constraints)²⁷ limits to growth. The complex nature of climate change makes predicting the external drivers to growth hard to quantify at the individual level, as these drivers affect communities as a whole. Drought, spring frost and heat waves are commonly mentioned as the main extreme events that could limit tree growth under climate change^{28,29,30,31,32,33,34,35}. To better understand these mechanisms, experiments are paramount to robustly tease apart the external vs internal drivers (e.g. warmer springs from severe drought later in the season—a common co-occurring reality in natural environments)^{36,37}. This is essential to refine forest carbon sequestration projections^{22,38}. However, experiments are most often performed on juvenile trees, which are critical for their role in forest regeneration projections, but their responses can hardly be translated to mature trees, which hold the overwhelming carbon biomass proportion of forests^{39,23,40}. To investigate how growing season shifts impact mature trees in their natural environments, unmanned aerial vehicle (UAV) imagery paired with machine learning has the capacity to acquire huge sample sizes, beyond what is achievable via traditional observational ground work, and at a better spatial and temporal resolution than satellites^{41,42,43}. Thus, I propose to use a combination of two experiments to test internal (Chapter 1) and external (Chapter 2) limits to growth along with a large-scale mixed-forest observational data project using UAV imagery and machine learning (Chapter 3). This will allow me to address the paradox of the absence of increased growth despite apparently improved growing season conditions.

Chapter 1. Continuation of the extended growing season experiment: Shifts in phenological phases have consequences for growth during the current growing season—a hypothesis I tested previously in my thesis—but experiments to date have not quantified whether longer seasons have lagging effects over subsequent years^{44,45,46,47,48}. My objective for this first chapter is to complete an additional project I conducted during my MSc but deliberately set aside from my thesis. In fall 2025, I completed a large-scale growing season length experiment using 15 replicates of seven species over two consecutive years, assessing different tree species' potential to stretch their activity schedules and whether this translates into increased growth over multiple growing seasons. In the first year of my proposed award tenure, I will leverage the tens of thousands of observations I collected in 2024 and 2025, analysing these data using a sophisticated joint Bayesian hierarchical model. Developing this complex tree growth model will build on an ongoing collaboration with Dr. Charles Margossian, a statistician and expert in Bayesian inference. While mostly computational, this first chapter will allow me to robustly test whether tree growth is driven by multi-year conditions, providing the field with a crucial modelling tool to test this hypothesis in natural systems²³.

Chapter 2. Drought and spring frost experiment—external drivers: With climate change, not only will growing season length shift, but trees will also experience shifts in the timing of moisture deficits leading to drought stress, and increased late-spring frosts⁴⁹. Tree-ring research shows that both droughts and frosts can result in important tissue loss^{50,51,52}. However, it is unclear whether trees exposed to droughts and spring frosts also grow less^{53,51,54,55}. Here, my objective will be to investigate how these two abiotic drivers affect tree growth. For this, I will conduct an experiment that I will start during the second year of the award tenure that consists of three drought treatments and an additional two spring frost treatments. I will use 15 replicates of 16 deciduous North American tree species (8 congeneric pairs to avoid potential confounding effects of shared evolutionary history), spanning different life history strategies, for all five treatments and a control, summing a total of 1080 individuals (a sample size consistent with my successful experiment that was completed during my MSc). For the drought treatments, I will move the trees to growth chambers at a warmer temperature and lower air humidity than ambient conditions to maximize evapotranspiration rates until they reach their respective wilting point (values at which soil water is not extractable by the plant). Then the trees will be moved back to ambient conditions and at constant irrigation. The three drought treatments will differ in their timing of occurrence to test the importance of drought timing. Thus, the first treatment will be conducted just after leaf-out; the second, one week before the solstice—period of peak growth for a lot of species^{56,57,58} (CITEVVDM); the last will happen near the end of the season, just before growth cessation. For spring frost treatments, I will place the trees in growth chambers early in the season at warm temperatures to trigger budburst. When the trees start to burst, I will place the first treatment for one hour in freezing growth chambers. For the second spring frost treatment, I will wait for the leaves to be fully elongated and then place the trees under the same freezing conditions as the first treatment⁵⁹. Both treatments will follow successful methods in Chamberlain *et al.* (2021). For all treatments, including the control, I will monitor phenology throughout the growing season, estimate biomass at the start and end of the growing season—using allometric equations, and I will equip a subset of trees with magnetic dendrometers that will provide valuable insight into the exact timing of changing growth in response to treatments.

Chapter 3. Growth timing×phenology drone imagery: An improved understanding of the differences in growth synchronicity with leaf phenology across species is critical in forest ecology, yet this relationship has received little attention^{60,61}. Thus, for the three years of my award tenure, I will launch a large-scale project using cutting-edge UAVs equipped with multispectral sensors×artificial intelligence technologies^{62,43,63} to gather a large amount of data on tree growth onset and end from a mixed-forest community. My objectives are to increase the accuracy of the data used by carbon cycle models—paramount in the context of rapid climate change^{24,25}—and improve the field of phenology remote sensing by challenging novel methods and building on them. The work will take place at a research station in St-Hypolyte (Qc) for three consecutive growing seasons. Using this site will allow me to follow up on work previously done by my laboratory¹⁷ as well as create a partnership with Dr. Etienne Laliberté, whose laboratory currently uses this site for their research⁶⁴. I will perform high-frequency UAV overflights to capture leaf phenology of every canopy tree throughout the growing season. Then I will use BalSAM, a promising model to accurately and efficiently segment tree crowns from repeated UAV images⁴³. This will allow me to gather a large amount of highly precise data from single trees within the forest community⁴³. With this data, I will be able to infer the start and end of the growing season for each species and individual within this forest community^{41,65}. Then, I will deploy 50 dendrometers distributed across the site to record stem diameter changes on canopy trees. Using high-resolution data across space and time will allow me to robustly infer a relationship between leaf phenology and growth seasonality.

Outreach: Given the widespread impacts of climate change on ecosystems, understanding how forest communities respond to prolonged growing seasons is crucial. Observing the responses of deciduous tree

species to extended seasons may reveal potential benefits for some species and harm for others. These shifts are likely to influence forest dynamics across North America, with potential feedback with future climate change. Therefore, using two experiments and a large-scale remote sensing project, I aim to understand how the growth dynamics of North American tree species will change with longer growing seasons.

References

1. Parmesan, C., Ryrholm, N., Stefanescu, C., Hill, J. K., Thomas, C. D., Descimon, H., Huntley, B., Kaila, L., Kullberg, J., Tammaru, T., Tennent, W. J., Thomas, J. A., and Warren, M. Poleward shifts in geographical ranges of butterfly species associated with regional warming. *Nature* **399**(6736), 579–583, June (1999).
2. Rosenzweig, C., Karoly, D., Vicarelli, M., Neofotis, P., Wu, Q., Casassa, G., Menzel, A., Root, T. L., Estrella, N., Seguin, B., Tryjanowski, P., Liu, C., Rawlins, S., and Imeson, A. Attributing physical and biological impacts to anthropogenic climate change. *Nature* **453**(7193), 353–357, May (2008).
3. Parmesan, C. and Yohe, G. A globally coherent fingerprint of climate change impacts across natural systems. *Nature* **421**(6918), 37–42, January (2003).
4. Cleland, E., Chuine, I., Menzel, A., Mooney, H., and Schwartz, M. Shifting plant phenology in response to global change. *Trends in Ecology & Evolution* **22**(7), 357–365, July (2007).
5. Lieth, H., Jacobs, J., Lange, O. L., Olson, J. S., and Wieser, W., editors. *Phenology and Seasonality Modeling*, volume 8 of *Ecological Studies*. Springer Berlin Heidelberg, Berlin, Heidelberg, (1974).
6. Woolway, R. I., Sharma, S., Weyhenmeyer, G. A., Debolskiy, A., Golub, M., Mercado-Bettín, D., Perroud, M., Stepanenko, V., Tan, Z., Grant, L., Ladwig, R., Mesman, J., Moore, T. N., Shatwell, T., Vanderkelen, I., Austin, J. A., DeGasperi, C. L., Dokulil, M., La Fuente, S., Mackay, E. B., Schladow, S. G., Watanabe, S., Marcé, R., Pierson, D. C., Thiery, W., and Jennings, E. Phenological shifts in lake stratification under climate change. *Nature Communications* **12**(1), 2318, April (2021).
7. Menzel, A., Sparks, T. H., Estrella, N., Koch, E., Aasa, A., Ahas, R., Alm-Kübler, K., Bissolli, P., Braslavská, O., Briede, A., Chmielewski, F. M., Crepinsek, Z., Curnel, Y., Defila, C., Donnelly, A., Filella, Y., Jatzak, K., Mestre, A., Peñuelas, J., Pirinen, P., Scheifinger, H., Striz, M., Susnik, A., Van Vliet, A. J. H., Wielgolaski, F., Zach, S., and Züst, A. European phenological response to climate change matches the warming pattern. *Global Change Biology* **12**(10), 1969–1976, October (2006).
8. Duputié, A., Rutschmann, A., Ronce, O., and Chuine, I. Phenological plasticity will not help all species adapt to climate change. *Global Change Biology* **21**(8), 3062–3073, August (2015).
9. Wolfe, D. W., Schwartz, M. D., Lakso, A. N., Otsuki, Y., Pool, R. M., and Shaulis, N. J. Climate change and shifts in spring phenology of three horticultural woody perennials in northeastern USA. *International Journal of Biometeorology* **49**(5), 303–309, May (2005).
10. Chmielewski, F.-M. and Rötzer, T. Response of tree phenology to climate change across Europe. *Agricultural and Forest Meteorology* **108**(2), 101–112, June (2001).
11. Fu, Y. H., Piao, S., Op De Beeck, M., Cong, N., Zhao, H., Zhang, Y., Menzel, A., and Janssens, I. A. Recent spring phenology shifts in western Central Europe based on multiscale observations. *Global Ecology and Biogeography* **23**(11), 1255–1263, November (2014).
12. Chuine, I. Why does phenology drive species distribution? *Philosophical Transactions of the Royal Society B: Biological Sciences* **365**(1555), 3149–3160, October (2010).
13. Peñuelas, J. and Filella, I. Responses to a Warming World. *Science* **294**(5543), 793–795, October (2001). Publisher: American Association for the Advancement of Science.

14. Gallinat, A. S., Primack, R. B., and Wagner, D. L. Autumn, the neglected season in climate change research. *Trends in Ecology & Evolution* **30**(3), 169–176, March (2015).
15. Jeong, S. and Medvigy, D. Macroscale prediction of autumn leaf coloration throughout the continental United States. *Global Ecology and Biogeography* **23**(11), 1245–1254, November (2014).
16. Cooke, J. E. K., Eriksson, M. E., and Junttila, O. The dynamic nature of bud dormancy in trees: environmental control and molecular mechanisms. *Plant, Cell & Environment* **35**(10), 1707–1728, October (2012).
17. Flynn, D. F. B. and Wolkovich, E. M. Temperature and photoperiod drive spring phenology across all species in a temperate forest community. *New Phytologist* **219**(4), 1353–1362, September (2018).
18. Körner, C. and Basler, D. Phenology Under Global Warming. *Science* **327**(5972), 1461–1462, March (2010). Publisher: American Association for the Advancement of Science.
19. Delpierre, N., Vitasse, Y., Chuine, I., Guillemot, J., Bazot, S., Rutishauser, T., and Rathgeber, C. B. K. Temperate and boreal forest tree phenology: from organ-scale processes to terrestrial ecosystem models. *Annals of Forest Science* **73**(1), 5–25, March (2016).
20. Keenan, T. F., Gray, J., Friedl, M. A., Toomey, M., Bohrer, G., Hollinger, D. Y., Munger, J. W., O’Keefe, J., Schmid, H. P., Wing, I. S., Yang, B., and Richardson, A. D. Net carbon uptake has increased through warming-induced changes in temperate forest phenology. *Nature Climate Change* **4**(7), 598–604, July (2014).
21. Dow, C., Kim, A. Y., D’Orangeville, L., Gonzalez-Akre, E. B., Helcoski, R., Herrmann, V., Harley, G. L., Maxwell, J. T., McGregor, I. R., McShea, W. J., McMahon, S. M., Pederson, N., Tepley, A. J., and Anderson-Teixeira, K. J. Warm springs alter timing but not total growth of temperate deciduous trees. *Nature* **608**(7923), 552–557, August (2022).
22. Green, J. K. and Keenan, T. F. The limits of forest carbon sequestration. *Science* **376**(6594), 692–693, May (2022).
23. Silvestro, R., Zeng, Q., Buttò, V., Sylvain, J.-D., Drolet, G., Mencuccini, M., Thiffault, N., Yuan, S., and Rossi, S. A longer wood growing season does not lead to higher carbon sequestration. *Scientific Reports* **13**(1), 4059, March (2023).
24. Richardson, A. D., Keenan, T. F., Migliavacca, M., Ryu, Y., Sonnentag, O., and Toomey, M. Climate change, phenology, and phenological control of vegetation feedbacks to the climate system. *Agricultural and Forest Meteorology* **169**, 156–173, February (2013).
25. Swidrak, I., Schuster, R., and Oberhuber, W. Comparing growth phenology of co-occurring deciduous and evergreen conifers exposed to drought. *Flora - Morphology, Distribution, Functional Ecology of Plants* **208**(10-12), 609–617, December (2013).
26. Kolář, T., Giagli, K., Trnka, M., Bednářová, E., Vavřík, H., and Rybníček, M. Response of the leaf phenology and tree-ring width of European beech to climate variability. *Silva Fennica* **50**(2) (2016).
27. Zohner, C. M., Mirzaghali, L., Renner, S. S., Mo, L., Rebindaine, D., Bucher, R., Palouš, D., Vitasse, Y., Fu, Y. H., Stocker, B. D., and Crowther, T. W. Effect of climate warming on the timing of autumn leaf senescence reverses after the summer solstice. *Science* **381**(6653), July (2023).

28. Tyree, M. T. and Zimmermann, M. H. *Xylem Structure and the Ascent of Sap*. Springer Series in Wood Science. Springer Berlin Heidelberg, Berlin, Heidelberg, (2002).
29. Choat, B., Brodribb, T. J., Brodersen, C. R., Duursma, R. A., López, R., and Medlyn, B. E. Triggers of tree mortality under drought. *Nature* **558**(7711), 531–539, June (2018).
30. Li, Y., Zhang, W., Schwalm, C. R., Gentine, P., Smith, W. K., Ciais, P., Kimball, J. S., Gazol, A., Kannenberg, S. A., Chen, A., Piao, S., Liu, H., Chen, D., and Wu, X. Widespread spring phenology effects on drought recovery of Northern Hemisphere ecosystems. *Nature Climate Change* **13**(2), 182–188, February (2023).
31. Trenberth, K. E., Dai, A., Van Der Schrier, G., Jones, P. D., Barichivich, J., Briffa, K. R., and Sheffield, J. Global warming and changes in drought. *Nature Climate Change* **4**(1), 17–22, January (2014).
32. Change, I. P. O. C. Detection and Attribution of Climate Change: from Global to Regional. In *Climate Change 2013 – The Physical Science Basis*, 867–952. Cambridge University Press 1 edition, March (2014).
33. Chiang, F., Mazdiasni, O., and AghaKouchak, A. Evidence of anthropogenic impacts on global drought frequency, duration, and intensity. *Nature Communications* **12**(1), 2754, May (2021).
34. Polgar, C. A. and Primack, R. B. Leaf-out phenology of temperate woody plants: from trees to ecosystems. *New Phytologist* **191**(4), 926–941, September (2011).
35. Reinmann, A. B., Bowers, J. T., Kaur, P., and Kohler, C. Compensatory responses of leaf physiology reduce effects of spring frost defoliation on temperate forest tree carbon uptake. *Frontiers in Forests and Global Change* **6**, 988233, February (2023).
36. Morin, X., Roy, J., Sonié, L., and Chuine, I. Changes in leaf phenology of three European oak species in response to experimental climate change. *New Phytologist* **186**(4), 900–910, June (2010).
37. Primack, R. B., Laube, J., Gallinat, A. S., and Menzel, A. From observations to experiments in phenology research: investigating climate change impacts on trees and shrubs using dormant twigs. *Annals of Botany* **116**(6), 889–897, November (2015).
38. Cabon, A., Kannenberg, S. A., Arain, A., Babst, F., Baldocchi, D., Belmecheri, S., Delpierre, N., Guerrieri, R., Maxwell, J. T., McKenzie, S., Meinzer, F. C., Moore, D. J. P., Pappas, C., Rocha, A. V., Szejner, P., Ueyama, M., Ulrich, D., Vincke, C., Voelker, S. L., Wei, J., Woodruff, D., and Anderegg, W. R. L. Cross-biome synthesis of source versus sink limits to tree growth. *Science* **376**(6594), 758–761, May (2022).
39. Augspurger, C. K. and Bartlett, E. A. Differences in leaf phenology between juvenile and adult trees in a temperate deciduous forest. *Tree Physiology* **23**(8), 517–525, June (2003).
40. Vitasse, Y. Ontogenic changes rather than difference in temperature cause understory trees to leaf out earlier. *New Phytologist* **198**(1), 149–155, April (2013).
41. Berra, E. F., Gaulton, R., and Barr, S. Assessing spring phenology of a temperate woodland: A multiscale comparison of ground, unmanned aerial vehicle and Landsat satellite observations. *Remote Sensing of Environment* **223**, 229–242, March (2019).

42. Piao, S., Liu, Q., Chen, A., Janssens, I. A., Fu, Y., Dai, J., Liu, L., Lian, X., Shen, M., and Zhu, X. Plant phenology and global climate change: Current progresses and challenges. *Global Change Biology* **25**(6), 1922–1940 (2019).
43. Teng, M., Ouaknine, A., Laliberté, E., Bengio, Y., Rolnick, D., and Larochelle, H. Bringing SAM to new heights: Leveraging elevation data for tree crown segmentation from drone imagery, , June (2025). arXiv:2506.04970 [cs].
44. Chapin, F. S., Schulze, E., and Mooney, H. A. The Ecology and Economics of Storage in Plants. *Annual Review of Ecology and Systematics* **21**(1), 423–447, November (1990).
45. Landhäusser, S. M., Pinno, B. D., Lieffers, V. J., and Chow, P. S. Partitioning of carbon allocation to reserves or growth determines future performance of aspen seedlings. *Forest Ecology and Management* **275**, 43–51, July (2012).
46. Lawrence, B. T. and Melgar, J. C. Variable Fall Climate Influences Nutrient Resorption and Reserve Storage in Young Peach Trees. *Frontiers in Plant Science* **9** (2018).
47. Martens, L. A., Landhäusser, S. M., and Lieffers, V. J. First-year growth response of cold-stored, nursery-grown aspen planting stock. *New Forests* **33**(3), 281–295, May (2007).
48. Schott, K. M., Pinno, B. D., and Landhäusser, S. M. Premature shoot growth termination allows nutrient loading of seedlings with an indeterminate growth strategy. *New Forests* **44**(5), 635–647, September (2013).
49. Dox, I., Mariën, B., Zuccarini, P., Marchand, L. J., Prislan, P., Gričar, J., Flores, O., Gehrmann, F., Fonti, P., Lange, H., Peñuelas, J., and Campioli, M. Wood growth phenology and its relationship with leaf phenology in deciduous forest trees of the temperate zone of Western Europe. *Agricultural and Forest Meteorology* **327**, 109229, December (2022).
50. Kramer, K., Bijlsma, R.-J., Hickler, T., and Thuiller, W. Why Would Plant Species Become Extinct Locally If Growing Conditions Improve? *International Journal of Biological Sciences* **8**(8), 1121–1129 (2012).
51. Baumgarten, F., Gessler, A., and Vitasse, Y. No risk—no fun: Penalty and recovery from spring frost damage in deciduous temperate trees. *Functional Ecology* **37**(3), 648–663, March (2023).
52. D’Andrea, E., Rezaie, N., Battistelli, A., Gavrichkova, O., Kuhlmann, I., Matteucci, G., Moscatello, S., Proietti, S., Scartazza, A., Trumbore, S., and Muhr, J. Winter’s bite: beech trees survive complete defoliation due to spring late-frost damage by mobilizing old C reserves. *New Phytologist* **224**(2), 625–631, October (2019).
53. Chamberlain, C. J. and Wolkovich, E. M. Late spring freezes coupled with warming winters alter temperate tree phenology and growth. *New Phytologist* **231**(3), 987–995, August (2021).
54. Lian, X., Piao, S., Li, L. Z. X., Li, Y., Huntingford, C., Ciais, P., Cescatti, A., Janssens, I. A., Peñuelas, J., Buermann, W., Chen, A., Li, X., Myneni, R. B., Wang, X., Wang, Y., Yang, Y., Zeng, Z., Zhang, Y., and McVicar, T. R. Summer soil drying exacerbated by earlier spring greening of northern vegetation. *Science Advances* **6**(1), eaax0255, January (2020).

55. Zhang, J., Gou, X., Alexander, M. R., Xia, J., Wang, F., Zhang, F., Man, Z., and Pederson, N. Drought limits wood production of *Juniperus przewalskii* even as growing seasons lengthens in a cold and arid environment. *CATENA* **196**, 104936, January (2021).
56. Anderson-Teixeira, K. J., Herrmann, V., Banbury Morgan, R., Bond-Lamberty, B., Cook-Patton, S. C., Ferson, A. E., Muller-Landau, H. C., and Wang, M. M. H. Carbon cycling in mature and regrowth forests globally. *Environmental Research Letters* **16**(5), 053009, May (2021).
57. D'Orangeville, L., Maxwell, J., Kneeshaw, D., Pederson, N., Duchesne, L., Logan, T., Houle, D., Arseneault, D., Beier, C. M., Bishop, D. A., Druckenbrod, D., Fraver, S., Girard, F., Halman, J., Hansen, C., Hart, J. L., Hartmann, H., Kaye, M., Leblanc, D., Manzoni, S., Ouimet, R., Rayback, S., Rollinson, C. R., and Phillips, R. P. Drought timing and local climate determine the sensitivity of eastern temperate forests to drought. *Global Change Biology* **24**(6), 2339–2351, June (2018).
58. McMahon, S. M. and Parker, G. G. A general model of intra-annual tree growth using dendrometer bands. *Ecology and Evolution* **5**(2), 243–254, January (2015).
59. Zohner, C., Rockinger, A., and Renner, S. Increased autumn productivity permits temperate trees to compensate for spring frost damage. *The New phytologist* **221** 2, 789–795 (2018).
60. Klein, T., Vitasse, Y., and Hoch, G. Coordination between growth, phenology and carbon storage in three coexisting deciduous tree species in a temperate forest. *Tree Physiology* **36**(7), 847–855, July (2016).
61. Kramer, K., Leinonen, I., and Loustau, D. The importance of phenology for the evaluation of impact of climate change on growth of boreal, temperate and Mediterranean forests ecosystems: an overview. *International Journal of Biometeorology* **44**(2), 67–75, August (2000).
62. Ball, J. G. C., Hickman, S. H. M., Jackson, T. D., Koay, X. J., Hirst, J., Jay, W., Archer, M., Aubry-Kientz, M., Vincent, G., and Coomes, D. A. Accurate delineation of individual tree crowns in tropical forests from aerial RGB imagery using Mask R-CNN. *Remote Sensing in Ecology and Conservation* **9**(5), 641–655, October (2023).
63. Ulku, I., Akagunduz, E., and Ghamisi, P. Deep Semantic Segmentation of Trees Using Multispectral Images. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* **15**, 7589–7604 (2022).
64. Cloutier, M., Germain, M., and Laliberté, E. Influence of temperate forest autumn leaf phenology on segmentation of tree species from UAV imagery using deep learning. *Remote Sensing of Environment* **311**, 114283, September (2024).
65. Fawcett, D., Bennie, J., and Anderson, K. Monitoring spring phenology of individual tree crowns using drone-acquired NDVI data. *Remote Sensing in Ecology and Conservation* **7**(2), 227–244, June (2021).